

Speaker Identification Challenge

IAAA Competition 2026: Speaker Identification

Open-Set Speaker Classification Challenge

Overview

In this challenge, you will build models that classify short audio chunks by speaker identity, while explicitly handling out-of-distribution (OOD) speakers.

We provide audio from 1,000 distinct people with different accents and recording conditions. For each person:

Roughly 50% of their audio is used for training.

The remaining 50% is held out for evaluation.

We define:

446 “known” speakers: labeled by their specific unique speaker-id (a UUID string).

554 “unknown / OOD” speakers: all grouped into a single class labeled "unknown".

Participants must train models that classify each audio chunk into 447 classes:

"unknown" = unknown / OOD speaker

[speaker-id] = specific known speakers (e.g.,

3943d8f3-d820-44ff-aba6-23c796fda87b

)

The goal is twofold:

Achieve high accuracy on the known speakers (identifying their specific speaker-id).

Correctly assign OOD speakers (never seen as distinct identities) to the "unknown" class.

Problem Description

Task

Given a short audio segment (chunk), your model must output a probability distribution over 447 classes:

p_{unknown} – probability that the chunk belongs to the unknown (OOD) group.

$p_{\text{id1}}, p_{\text{id2}}, \dots, p_{\text{id446}}$ – probabilities for each of the 446 known speaker-ids.

For evaluation, we:

Convert your probabilities to a predicted label via

argmax

.

Compare those predicted labels to the ground-truth labels (either

"unknown"

or a specific speaker-id string).

Compute the evaluation metric (detailed below).

Dataset

Speakers and splits

Total known speakers: 446

Total unknown/OOD speakers: 554

Total people: 1,000

For each speaker:

Audio is split into training and evaluation chunks, approximately 50/50:

Training chunks are provided to you.

Evaluation chunks are kept hidden by the organizers.

Training labels

For each training audio chunk, we provide a metadata CSV mapping the audio file to its label:

Speaker-ids: individual known speakers (represented as UUID strings).

"unknown": audio belonging to any of the 554 unknown speakers. All of these speakers are collapsed into this single class.

This means you can train your model as a standard 447-way classifier, where the

"unknown"

class is intentionally broad and heterogeneous: it represents “some speaker that is not one of the 446 known identities.”

Evaluation labels

On the evaluation set:

For audio from known speakers, labels remain their specific speaker-id.

For audio from unknown speakers, labels are "unknown".

You never see which of the OOD persons a specific

"unknown"

label corresponds to. The core challenge is to assign speaker-ids accurately, and group OOD speakers as "unknown".

File layout (example)

audio/

train/

e9105299-285b-4df7-8c66-e4b3b721e8c8.mp3

016a2324-dc96-4f59-8d96-fb387afb1fff.mp3

57e9178b-7153-4e43-8fe4-6aacea3c9118.mp3

ef43d054-a0a8-4ce3-9362-498cf58093b2.mp3

...

eval/ # hidden evaluation set (not shipped to participants)

...

metadata/

train_labels.csv

train_labels.csv

contains the following format:

audio_file,speaker_id

e9105299-285b-4df7-8c66-e4b3b721e8c8.mp3,3943d8f3-d820-44ff-aba6-23c796fda87b

016a2324-dc96-4f59-8d96-fb387afb1fff.mp3,845cbfc6-af92-4fa0-8b33-4e4e0e3b002a

57e9178b-7153-4e43-8fe4-6aacea3c9118.mp3,unknown

ef43d054-a0a8-4ce3-9362-498cf58093b2.mp3,6531a140-1e3f-40f4-b3be-7871fed19a85

Note: Exact metadata columns, paths, and directory nesting will be detailed in the competition “Data” section.

Evaluation Pipeline

The organizers will run your submitted model(s) directly on the hidden evaluation dataset in a controlled environment.

Evaluation Metrics

The primary scoring metric for this competition is the Macro-Averaged F1-score across all 447 classes.

Because the dataset contains 446 known speakers and 1 aggregated unknown class, using a macro-averaged F1-score ensures that the model is evaluated fairly. It prevents the model from achieving a high score simply by over-predicting the majority class, and strictly balances precision and recall across both the individual known speakers and the

"unknown"

OOD class.

Submission & Evaluation details (API, packaging, runtime limits): TO_BE_FILLED BY ORGANIZERS

Use of External Data

You may:

Use publicly available pretrained audio models.

Use external speech datasets.

Perform pretraining or self-supervised learning.

You must not:

Use labeled data containing evaluation speakers.

Access hidden evaluation labels.

Rules

Model behavior

Your model must produce a 447-way probability distribution for each chunk (classes

"unknown"

and 446 specific speaker-ids).

Ensembling

Ensembling multiple models is allowed.

Reproducibility

Top teams may be required to share training and architecture details.

Fair play

The organizers reserve the right to disqualify teams that violate these rules.